Loan Default Prediction

Project Summary

Project Goal

Build a machine learning model that predicts whether a loan application will be approved, using applicant details such as income, credit history, education, employment status, and property area. The model is packaged with its encoders and scaler and served through a Streamlit web app as a quick, internal screening aid for loan officers, not a replacement for full underwriting.

Chosen Model & Final Accuracy

Five algorithms were trained and tuned with **GridSearchCV** (5-fold cross validation): Logistic Regression, Random Forest, SVM, KNN, and XGBoost. **Logistic Regression** (C=0.01, solver='liblinear') gave the highest test accuracy and was selected as the final model.

80.04%	86.18%	0.84 / 0.99	0.96 / 0.58
Training Accuracy	Test Accuracy	Approved: Precision / Recall	Not Approved: Precision / Recall

Model	Train Acc.	Test Acc.
Logistic Regression (selected)	80.04%	86.18%
Random Forest	80.65%	85.37%
SVM	79.84%	85.37%
KNN	80.24%	85.37%
XGBoost	80.04%	85.37%

Insights from Exploratory Data Analysis

- **Credit history dominates approval outcomes:** applicants with a credit history were approved 79.0% of the time, versus only 7.9% for those without one — by far the strongest signal in the dataset.
- **Class imbalance:** 68.7% of applications in the training data were approved versus 31.3% not approved, which the model reflects in its higher recall for the approved class.
- **Applicant income alone is a weak predictor:** mean applicant income was nearly identical between approved (5,384) and not-approved (5,446) applicants, showing income by itself does not separate the two groups.
- **Property area matters moderately:** Semiurban applicants had the highest approval rate (76.8%), ahead of Urban (65.8%) and Rural (61.5%).
- **Education has a smaller effect:** Graduates were approved at 70.8% versus 61.2% for non-graduates.
- **Missing data** was present in 7 columns (most notably Credit_History, Self_Employed, and LoanAmount) and was imputed using the mode for categorical fields and the median for numerical fields before modeling.